



Oxygen Gas Sensor O2/M-1

O2 Gas Sensor in Miniature Housing

Key Features

- Fast equilibration time
- Long-Life sensor with no consumable parts and lead-free (RoHS compliant)

Applications

- Trace oxygen measurements
- Continuous Air Quality Monitoring
- Safety and Environmental Control

For further information about usage of Membrapor sensors, see application note [MEM1](#).

Measurement

Operation Principle	3-Electrode Electrochemical
Nominal Range	0 - 1 %
Maximum Overload	21 %
Inboard Filter	-
Output Signal	1300 ± 300 µA/%
Resolution (Electronics dependent)	< 50.0 ppm
T90 Response Time	< 10 s
Typical Baseline Range (pure N2, 20°C)	-0.03 % to 0.01 %
Maximum Zero Shift (+20°C to +40°C)	see Graph
Repeatability	< 2 % of signal
Output Linearity	Linear
Gain (Only applies to 4-Electrode sensors)	-

Performance data recorded at 20 – 25 °C, 30 - 50% RH, 900 - 1100 mbar

The data contained in this document is for guidance only. Membrapor AG accepts no liability for any consequential losses, injury or damage resulting from the use of this document or from any omissions or errors herein. Customers should test under their own conditions to ensure that the sensors are suitable for their own requirements.



Oxygen Gas Sensor O2/M-1

Electrical

Rec. Load Resistor	10 - 33 Ω
Bias (V_Sens-V_Ref)	-600 mV
Conformity to RoHS directive	RoHS Compliance

Environmental

Relative Humidity Range	15 % to 90 % RH non-condensing
Temperature Range	-40 °C to 50 °C
Pressure Range	Atmospheric
Pressure Coefficient	N.D.
Humidity Effect	None

Lifetime

Expected Operation Life	5+ years in air
Expected Long Term Output Drift	< 4% signal per 3 years
Filter Life	not applicable
Storage Life	6 months in container
Recommended Storage Temperature	-10°C - 30°C
Warranty Period	3 years from date of dispatch

Performance data recorded at 20 – 25 °C, 30 - 50% RH, 900 - 1100 mbar

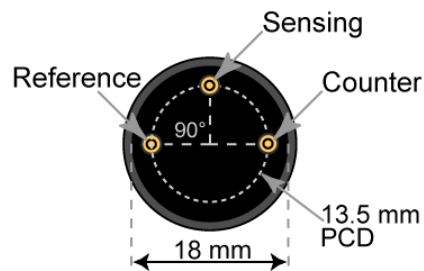
The data contained in this document is for guidance only. Membrapor AG accepts no liability for any consequential losses, injury or damage resulting from the use of this document or from any omissions or errors herein. Customers should test under their own conditions to ensure that the sensors are suitable for their own requirements.



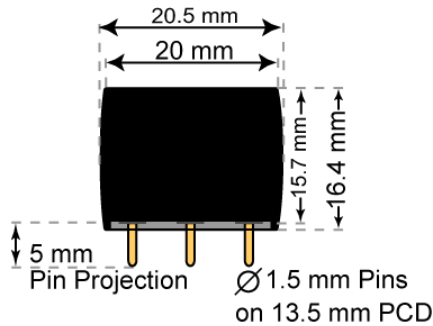
Oxygen Gas Sensor O2/M-1

Miniature-Size Outline Dimensions

BOTTOM VIEW



SIDE VIEW



± 0.10 mm

Mechanical

Weight	5.5 g
Orientation	Any
Housing material	Polycarbonate

Performance data recorded at 20 – 25 °C, 30 - 50% RH, 900 - 1100 mbar

The data contained in this document is for guidance only. Membrapor AG accepts no liability for any consequential losses, injury or damage resulting from the use of this document or from any omissions or errors herein. Customers should test under their own conditions to ensure that the sensors are suitable for their own requirements.



Oxygen Gas Sensor O2/M-1

Cross Sensitivity Data

The table below does not claim to be complete. We recommend using the target gas for calibration purposes. Using surrogate (interfering) gases can result in inaccuracies in the final calibration. Please contact Membrapor AG for further support regarding cross sensitivities.

Interfering Gas	Cross-Sens. [%]
H ₂ S	N.D.
SO ₂	N.D.

Important Notes

- Conversion factor: 10000 ppm = 1%

Performance data recorded at 20 – 25 °C, 30 - 50% RH, 900 - 1100 mbar

The data contained in this document is for guidance only. Membrapor AG accepts no liability for any consequential losses, injury or damage resulting from the use of this document or from any omissions or errors herein. Customers should test under their own conditions to ensure that the sensors are suitable for their own requirements.

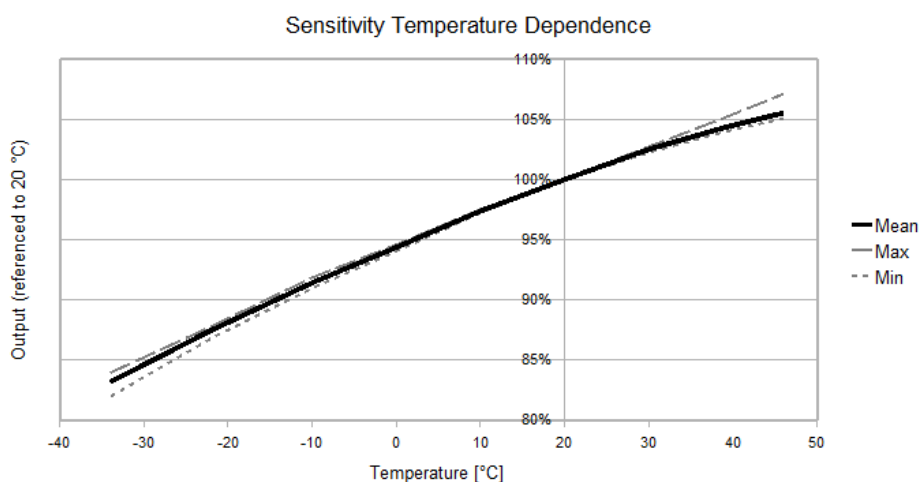


Oxygen Gas Sensor O2/M-1

Temperature dependence

The output of an electrochemical sensor varies with temperature. The graphs below show the temperature-dependent variation of baseline and sensitivity, respectively. The results shown here are raw data (batch average) without any post-processing steps. The sensitivity and baseline are referenced to the signal at 20°C (reference point).

Please note: It is highly recommended to acquire the temperature dependence curves with the whole instrument. The sampling system, the humidity, the electronics and the interaction between the electronics and the sensor have a significant impact on the temperature dependence of the final measurement reading.



Baseline shifted with respect to reference point at 20°C.

Performance data recorded at 20 – 25 °C, 30 - 50% RH, 900 - 1100 mbar

The data contained in this document is for guidance only. Membrapor AG accepts no liability for any consequential losses, injury or damage resulting from the use of this document or from any omissions or errors herein. Customers should test under their own conditions to ensure that the sensors are suitable for their own requirements.